Knee Joint Assessment and treatment

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Learning Outcomes

By the End of this lesson, you should be able to:

- Briefly review the anatomy and components of the knee joint.
- Practice the steps of MSK assessment relevant to the knee joint.

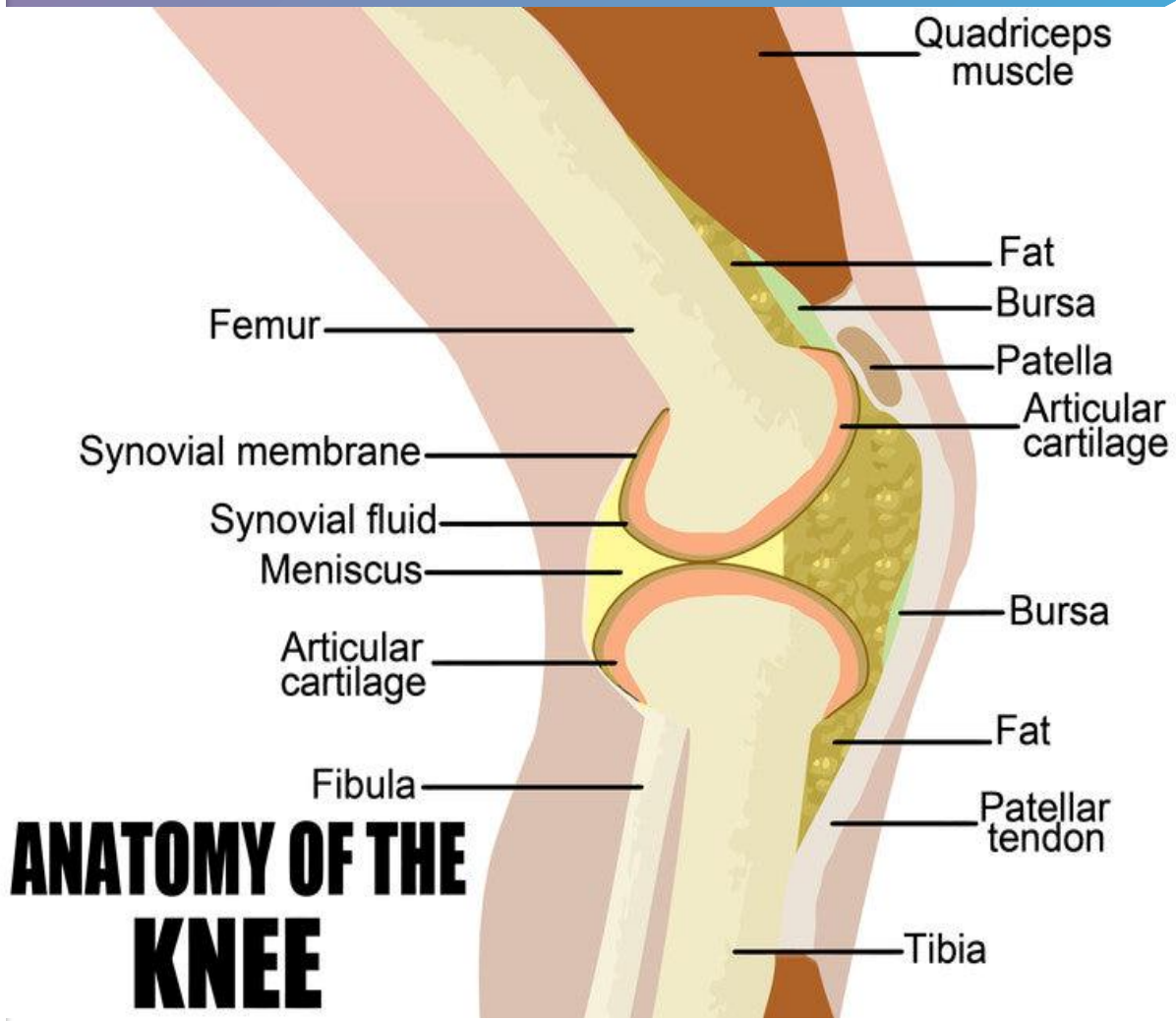


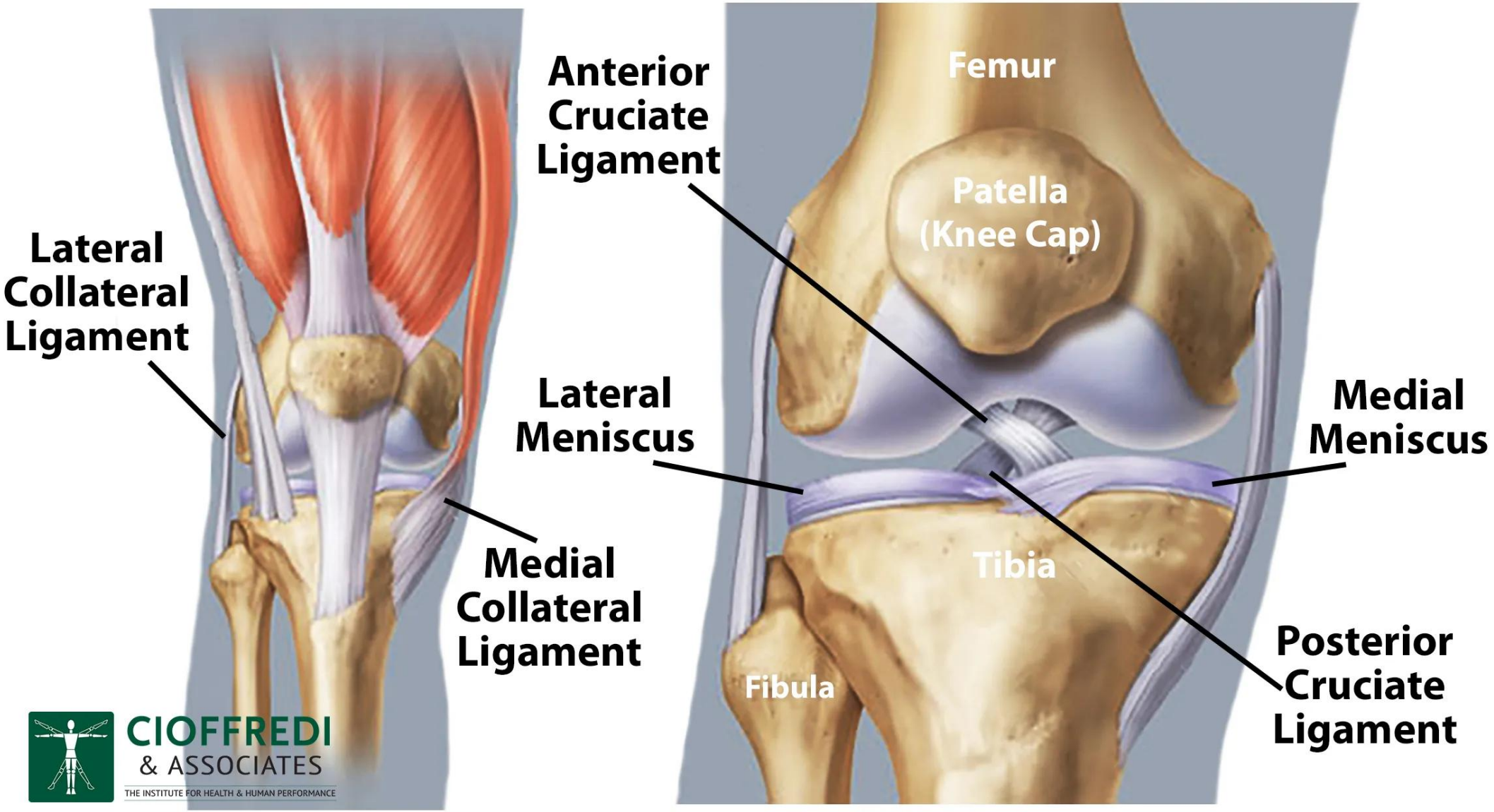
Assessment of the Knee Joint

- joint Anatomy, including composition, bones, ligaments, Muscles, bursae, menisci ...etc

- Subjective assessment: Tutorial

- Objective assessment




Lateral Collateral Ligament
Medial Collateral Ligament
Anterior Cruciate Ligament
Femur
Patella (Knee Cap)
Lateral Meniscus
Medial Meniscus
Tibia
Fibula
Posterior Cruciate Ligament
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Subjective Assessment

Mechanism of Injury?	On movement, is there any grating or clicking in the knee?
Trauma (Direct or indirect)?	Is the joint swollen? Does the swelling occur with activity or several hours after activity, or does the joint feel tight at rest?
Has the knee been injured before, or does it have any feeling of weakness?	Does the knee “give way?
Was there a “pop” when the injury occurred?	Has the knee ever locked?
Did the injury occur during acceleration, during deceleration, or when the patient was moving at a constant speed?	
Is there any pain? If so, where? What type is it? Retropatellar? Does the patient point to one spot with one finger or a more general area indicating the problem is more diffuse, aching?	

ACL

Pattern of injury



A twisting and hyperextension injury commonly causes an ACL +/- PCL tear.



A hyperextension injury in a nonweight bearing leg can cause an ACL or PCL tear.



A hyperflexion injury such as a fall in skiing causes an ACL tear.

PCL

Mechanism of Injury



Direct blow to
Anterior tibia



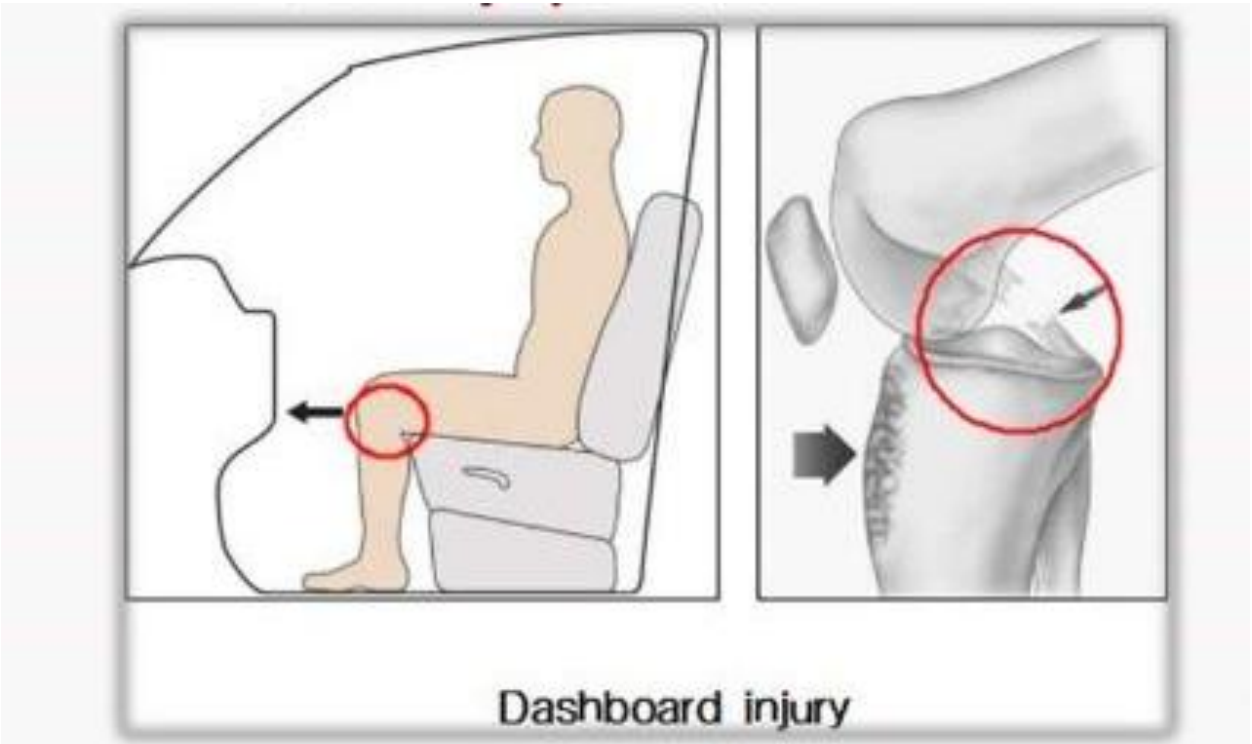
Hyperextension injury



Dashboard injury



Fall onto a flexed knee with foot
In plantarflexion



Objective Assessment

1- Observation

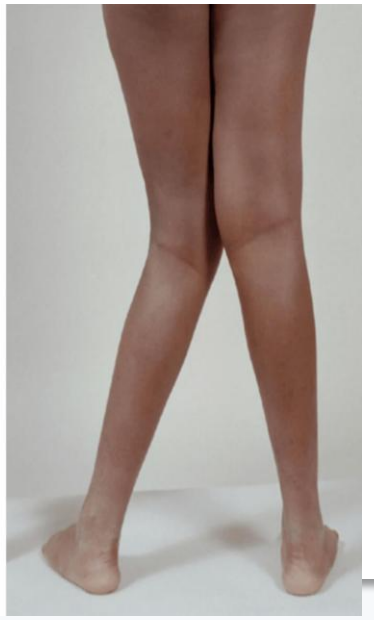
A. Informal: dynamic and static posture and movement, quality of movement…etc

B. Formal: 1. Posture: of the who lower limb, abnormal femoral rotation, abnormal tibial rotation or alignment.









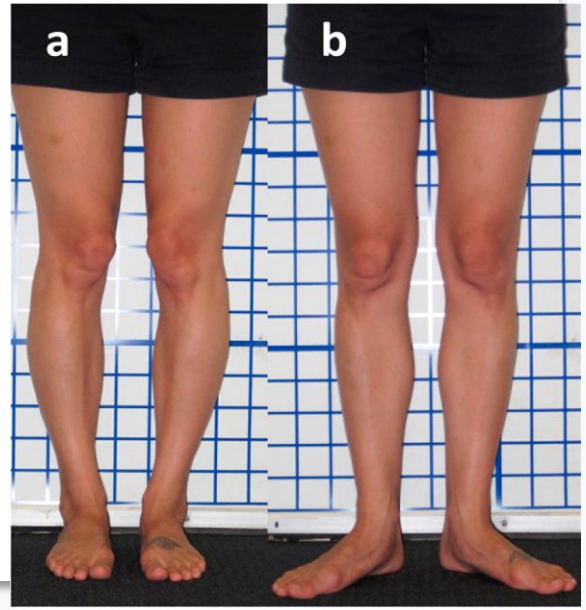
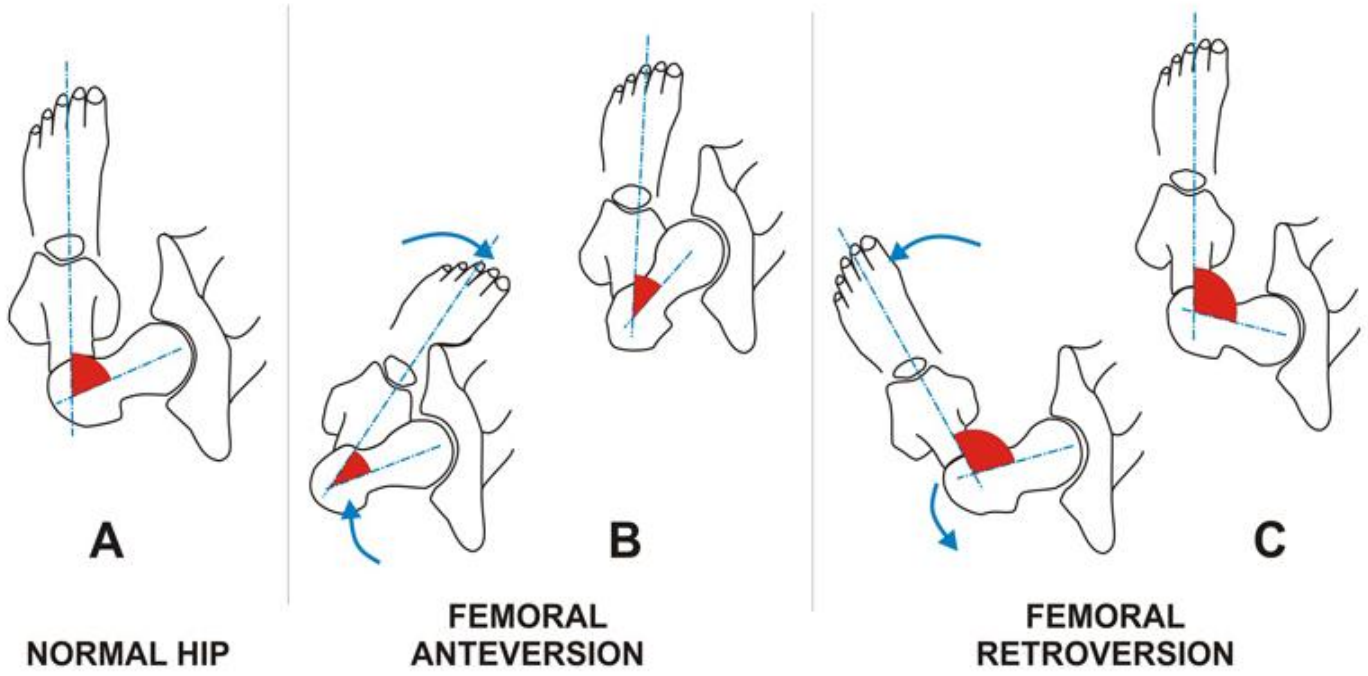


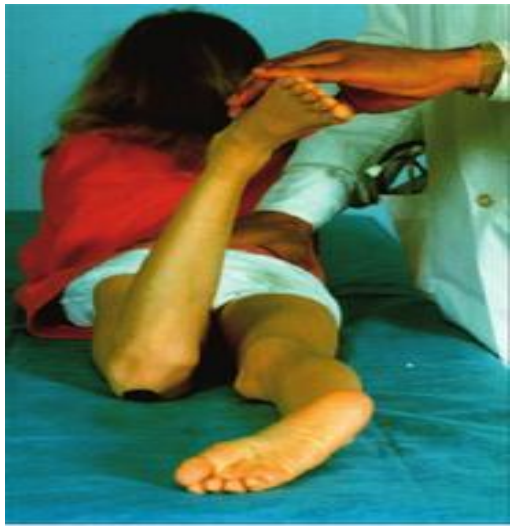
Fig. 2

1- Observation

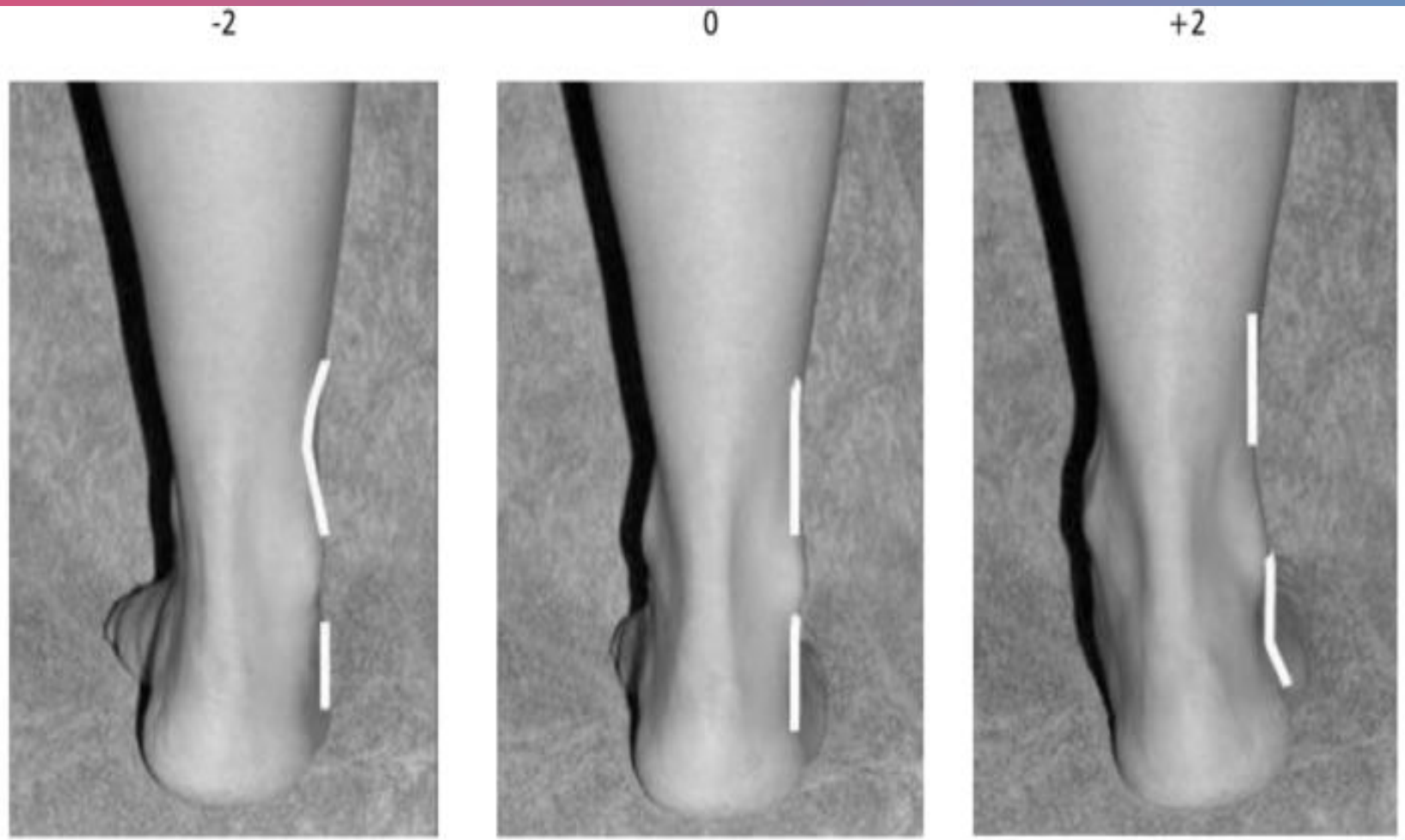


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1



-2	-1	0	+1	+2
Curve below the malleolus either straight or convex	Curve below the malleolus concave, but flatter/more shallow than the curve above the malleolus	Curve above and below the malleolus is identical	Curve below the malleolus more concave than curve above malleolus	Curve below the malleolus markedly more concave than curve above malleolus

Knee Joint assessment

1- OBSERVATION (CONT.)

2. Observation of Posture.

2. Observation of Muscle form.

3. Observation of soft tissue

4. Observation of Balance

5. Observation of Gait

2- ROM TESTING:

What are the movements to assess? What are you looking for?

- Active ROM testing: using Goniometer Flexion/Extension
- Passive ROM testing : over-pressure and endfeel?
- Accessory movement testing? Do you need to assess it? How will we use it? (Taught later)

Knee Joint assessment (Observation)

1. Observation of Posture.
2. Observation of Muscle
3. Observation of soft tissue
4. Observation of Balance
5. Observation of Gait

Muscle Tests

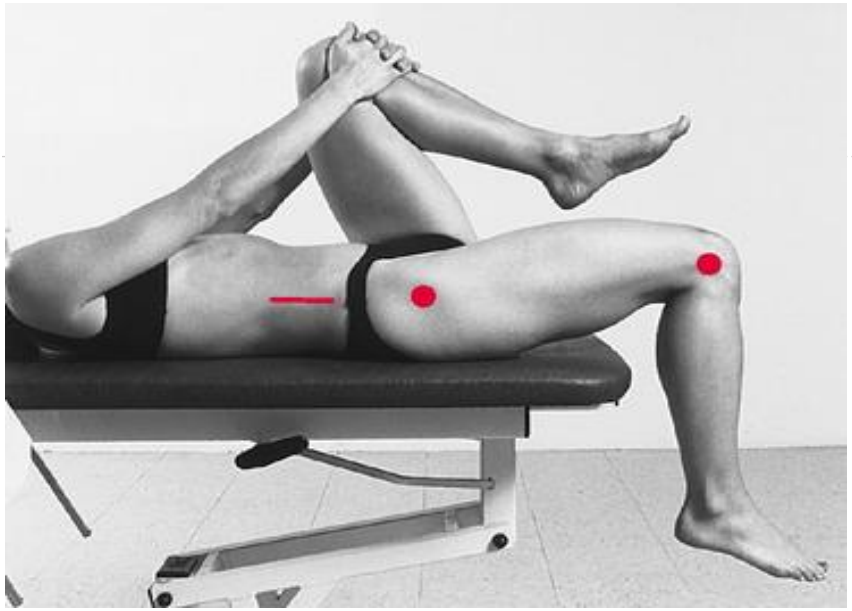
- 1- Muscle strength: Which muscles to test?
- Knee extension.
- Knee flexion in supine.
- Ankle dorsiflexion.
- Ankle plantar flexion.
- Knee flexion in prone.
- Knee flexion at 30° in sitting.

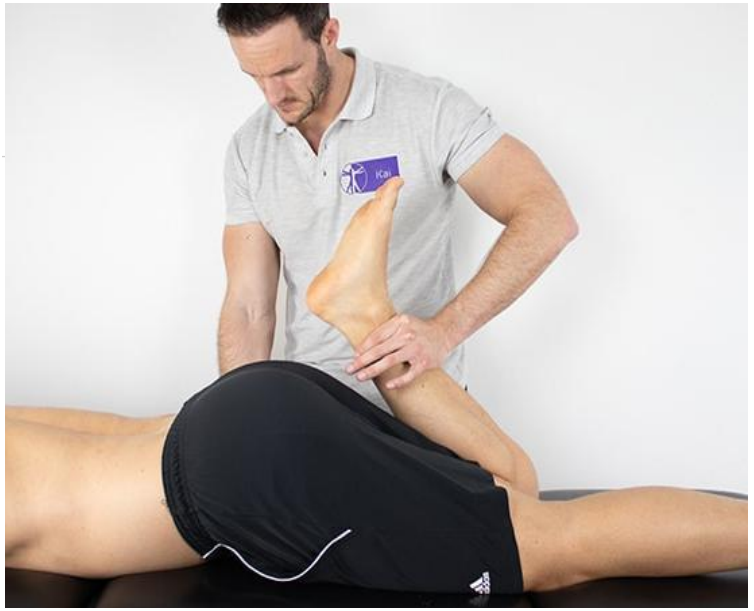


Muscle Tests

- 2- Muscle length (flexibility): What muscles to test or suspect their shortening.

(think beyond the knee)









Joint integrity tests Collateral Ligaments (Stress tests)

A. Ligaments Integrity:



1.Abduction (valgus) stress with knee in full extension

1. **Medial collateral ligament (superficial and deep fibers) (20-30)**
2. **Posterior oblique ligament (20-30)**
3. Posteromedial capsule
4. Anterior cruciate ligament
5. **Posterior cruciate ligament (20-30)**
6. Medial quadriceps expansion
7. Semimembranosus muscle

Joint integrity tests Collateral Ligaments (Stress tests)

A. Ligaments Integrity:



Adduction (varus) stress with knee in full extension

1. **Lateral collateral ligament (20-30 +external rotation of the tibia)**

2. Posterolateral capsule (20-30 +external rotation of the tibia)

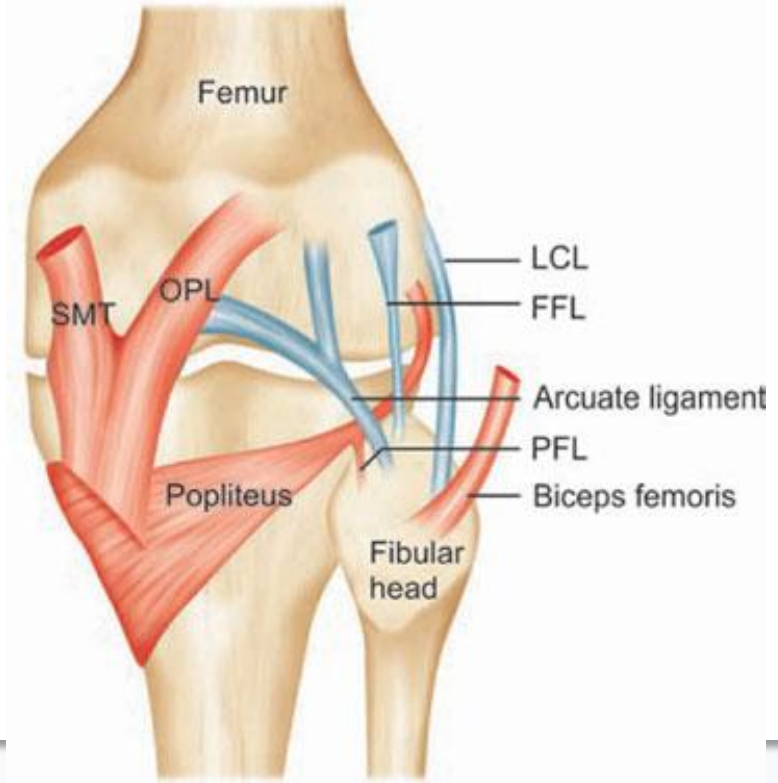
3. Arcuate-popliteus complex (20-30 +external rotation of the tibia)

4. Biceps femoris tendon

5. Anterior cruciate ligament

6. Posterior cruciate ligament

7. Lateral gastrocnemius muscle



Joint integrity tests 2- ACL and PCL

A. Ligaments Integrity:

1- Collateral Ligaments (Stress tests)





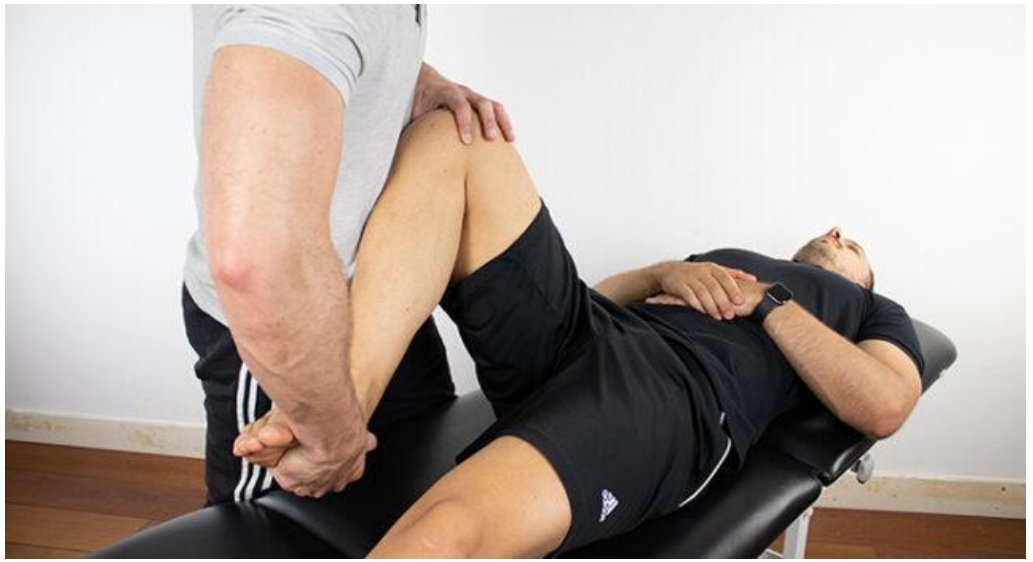




B- Meniscal integrity

- McMurray test
- Thessaly test
- Joint Line tenderness





Testing of Swelling of the knee



The patient is lying in supine with the leg extended. The physiotherapist puts pressure on the proximal side of the knee in an effort to squeeze the fluid out of the suprapatellar pouch.
The fluid can be moved under the patella while maintaining the pressure on the suprapatellar pouch; the therapist uses his/her other hand to press up on the medial and lateral recesses forcing the fluid under the patella . Tapping down the patella with the index to create an upward and downward movement and a palpable ‘click’ as the patella hits the underlying femur.
If the test is negative the femur and the patella are already in contact.



Bulge, wipe or stroke test / Fluid displacement test

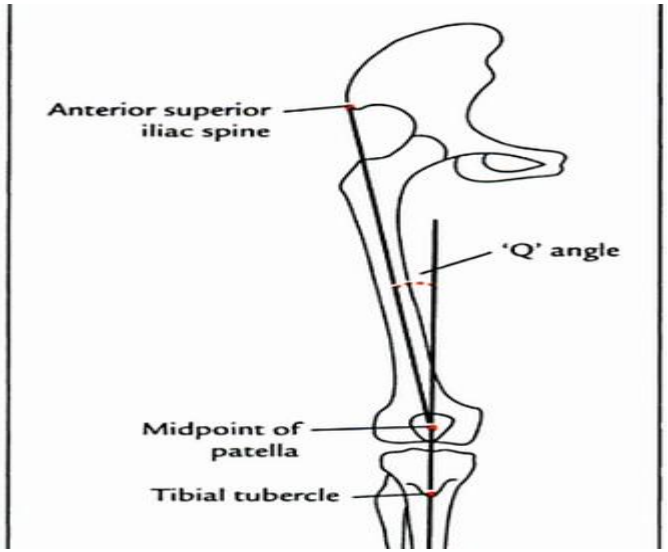
Patellofemoral joint assessment

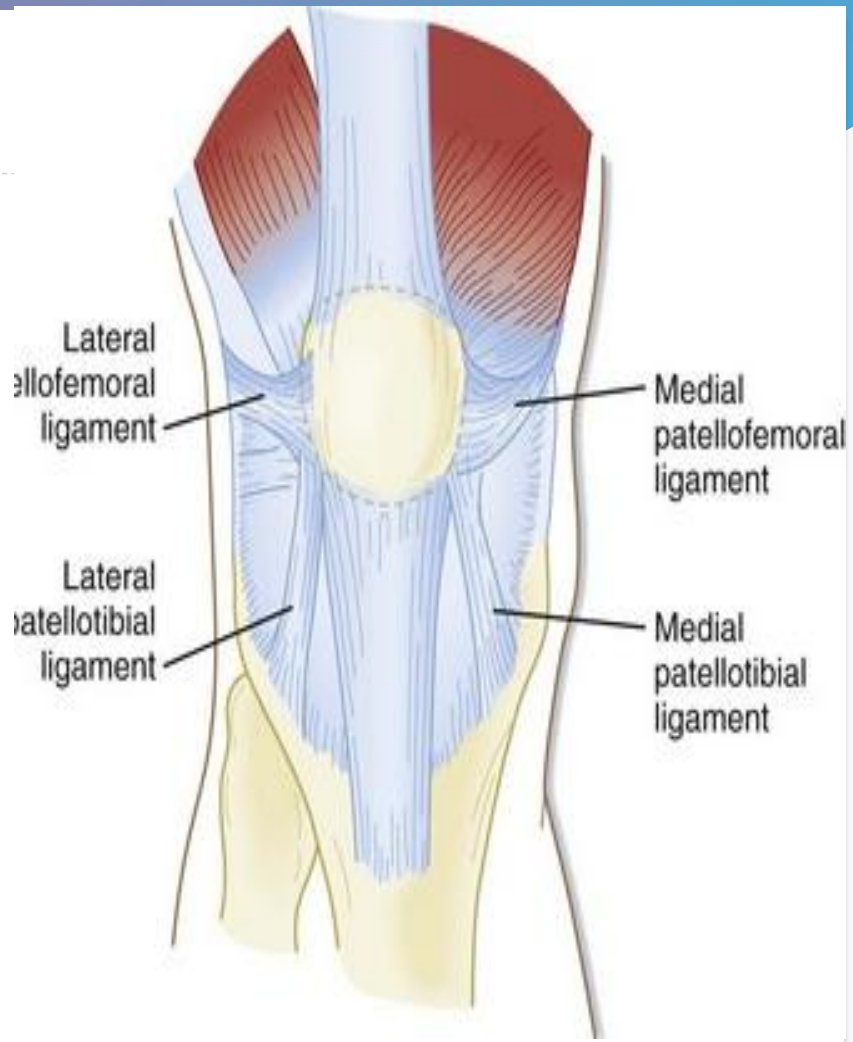
Conditions that may affect the PFJ: OA (Chondromalicia), mal-tracking- recurrent dislocation,

Related area conditions:

Hoffa's Pad, Osgood shlutter PFP syndrome

1. VMO strength
2. Clark's test
3. Firbank apprehension test
4. Palpation





Zohler’s (Clarke’s) sign for chondromalacia patellae.



Vastus medialis coordination test.

The patient is asked to slowly extend the knee without pressing into the examiner’s fist or lifting the leg away from the fist while trying to achieve full extension. The test is considered positive if the patient cannot fully extend the knee or has difficulty achieving full extension smoothly or tries to use the hip flexors or extensors to accomplish the task.



Patella maltracking sign

- The J-sign in patellar maltracking | BMJ Case Reports

- The J-sign in patellar maltracking

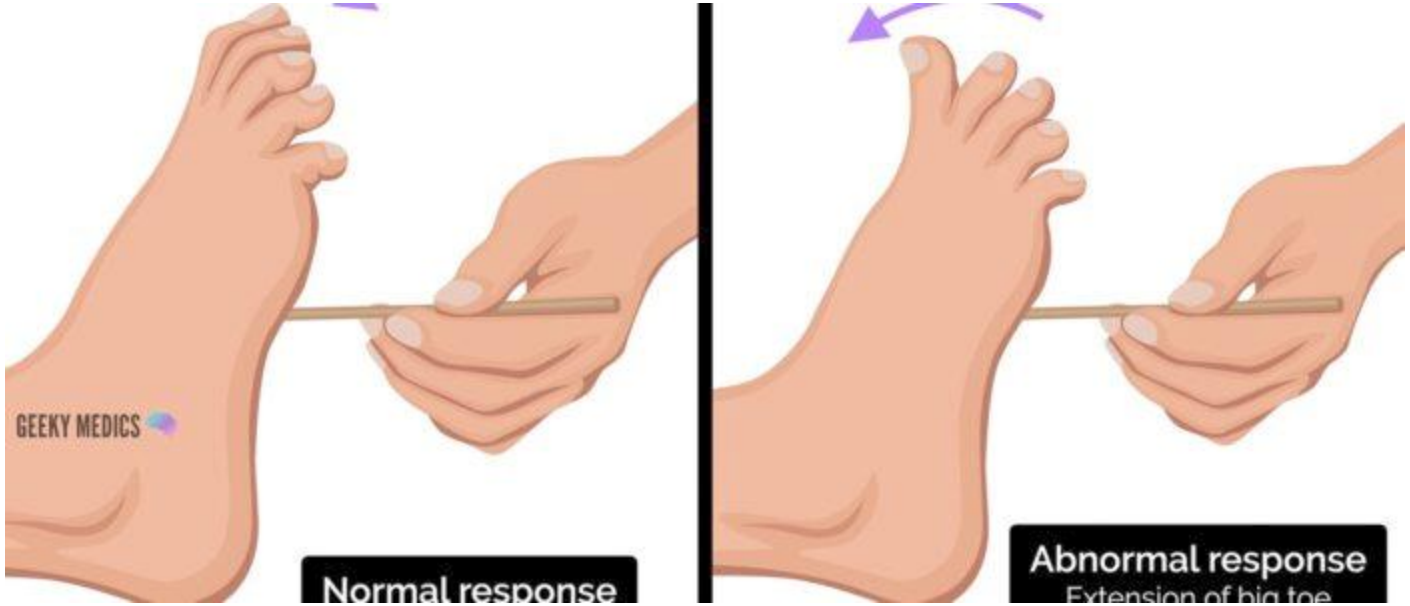
1. Omar Hadidi,
2. Prasad Ellanti,
3. Marc Lincoln,
4. Niall Hogan

- https://doi.org/10.1136/bcr-2017-222887

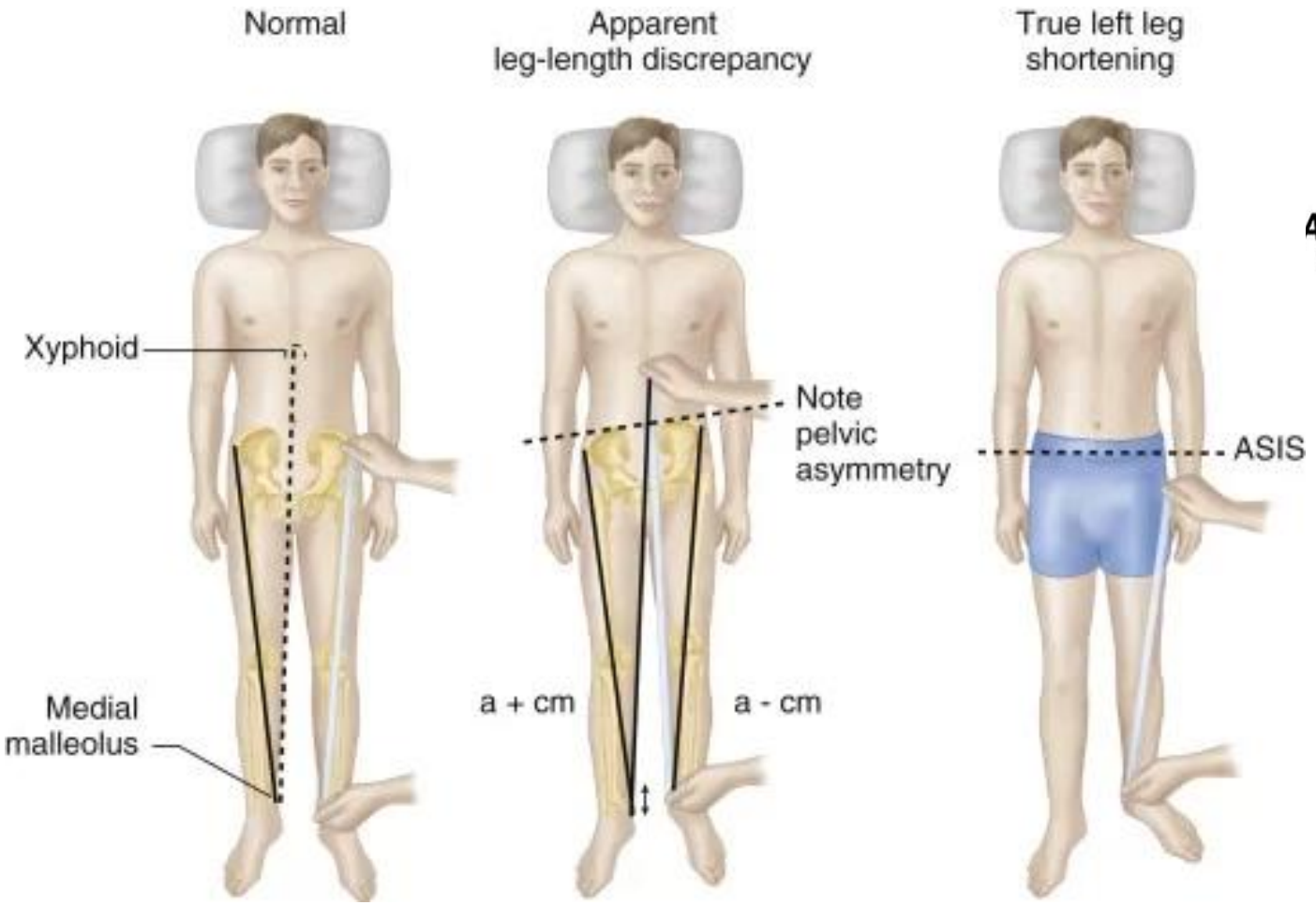


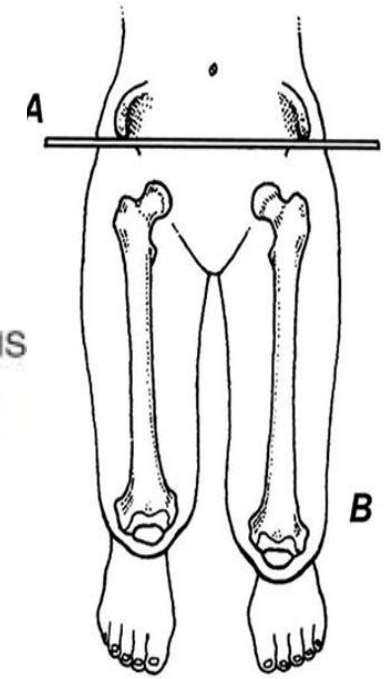
Neurological testing

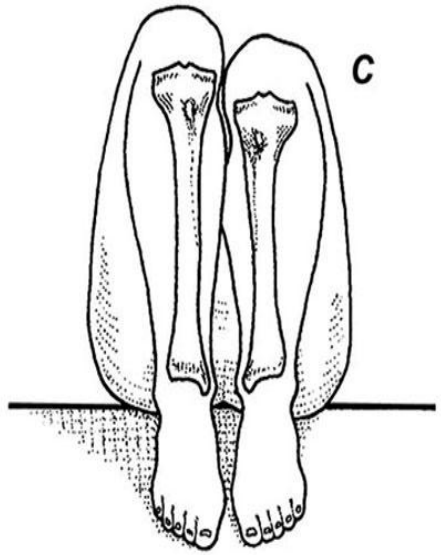
- Integrity of the nervous system: Dermatomes (Lumbar plexus) Myotomes (lumbar plexus) and reflexes (knee and ankle jerks) UMN screening (Babinski test, clonus)
- Dynamics of the nervous system: (Femoral nerve and Sciatic nerve)



Leg length discrepancy test







Functional testing of Single Leg Loading

	NWB thigh not held in neutral		
Knee position	Patella pointing towards 2nd toe (noticeable valgus)		
	Patella pointing past inside of foot (significant valgus)		
Steady stance	Touches down with NWB foot		
	Stance leg wobbles noticeably		

Functional testing of Single Leg Loading

QASLS		Optimal	Sub-optimal
Arm strategy	Excessive arm movement to balance		
Trunk alignment	Leaning in any direction		  
Pelvic plane	Loss of horizontal plane		 
	Excessive tilt or rotation		
Thigh motion	WB thigh moves into hip adduction		

Special tests related to cases

Hofffa’s pad test



Palpation

MEDIAL COLUMN

1A	- MCL injury
1B	- Medial retinaculum injury
	- Plica
	- Valgus ankle joint
1C	- Medial meniscus tear
	- Degenerative joint disease
	- Medial tibial plateau fracture
1D	- Pes anserine tendons
	- Pes anserine bursitis

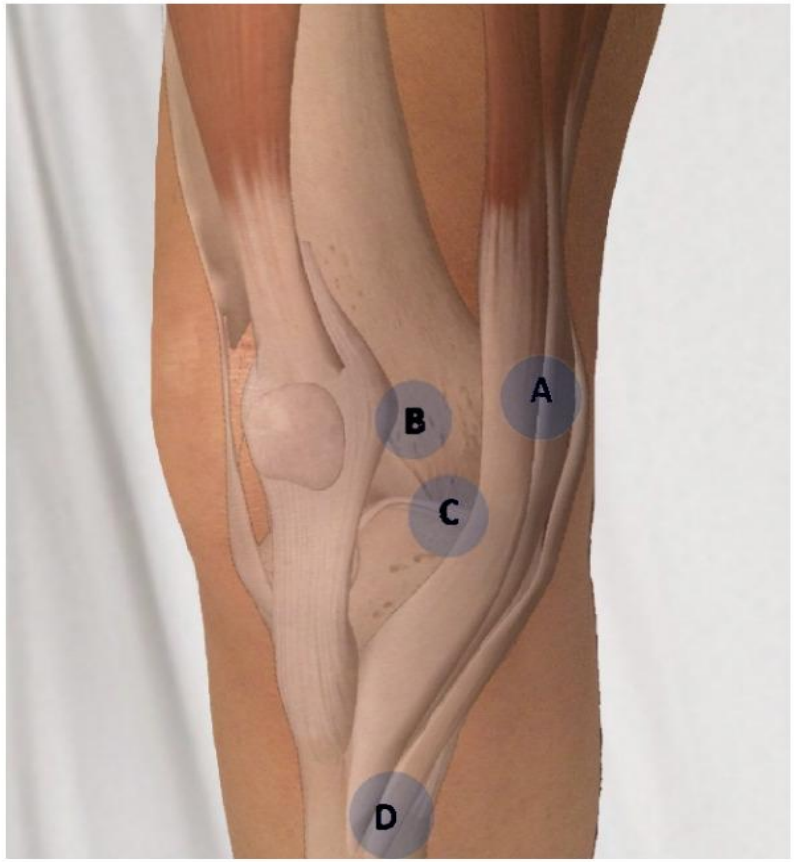


Figure 1 – Medial Column

MIDDLE COLUMN

2A - Quad tendonitis

2B - Patella fracture

- Patellar neuroma

- Prepatellar bursitis

2C - Patellar tendonitis

- Hoffa's disease

2D - Osgood-Schlatter's (adolescent)

- Tibial tubercle fracture (adults)

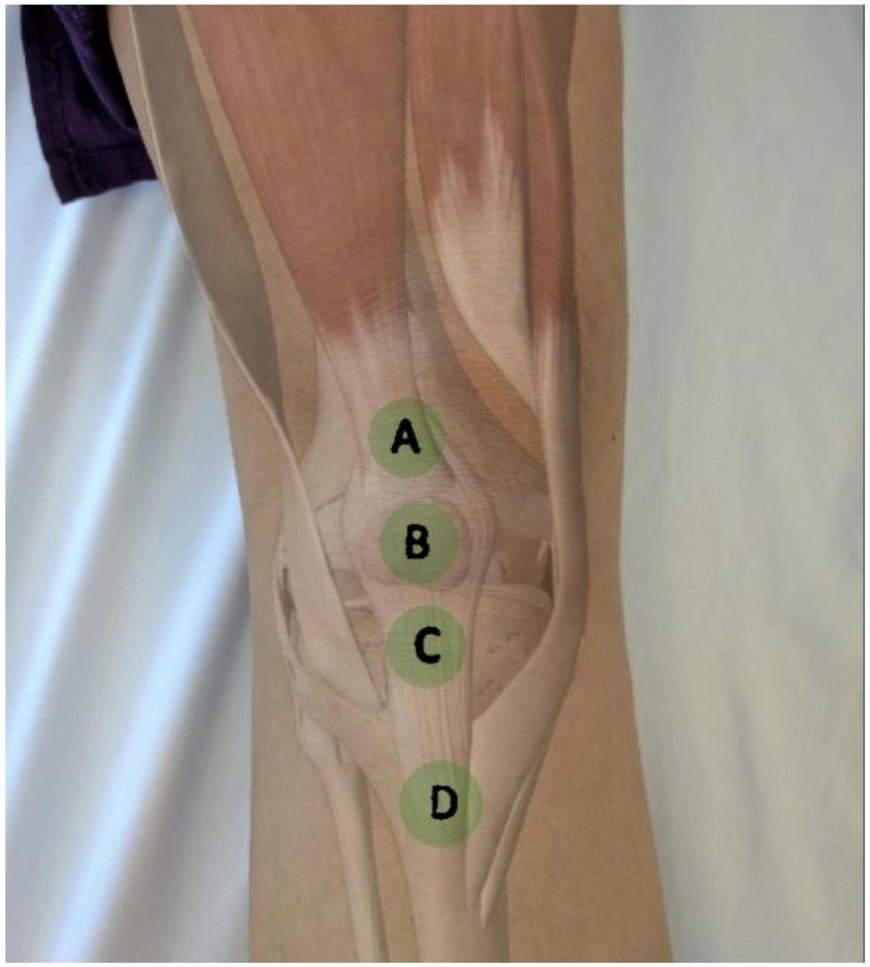


Figure 2 – Middle Column

LATERAL COLUMN

3A - IT band syndrome at the knee

3B - Patellar instability

3C - Lateral meniscus tear

- Degenerative joint disease

- Lateral tibial plateau fracture

3D - Proximal fibular fracture

- Proximal tibiofibular joint injury

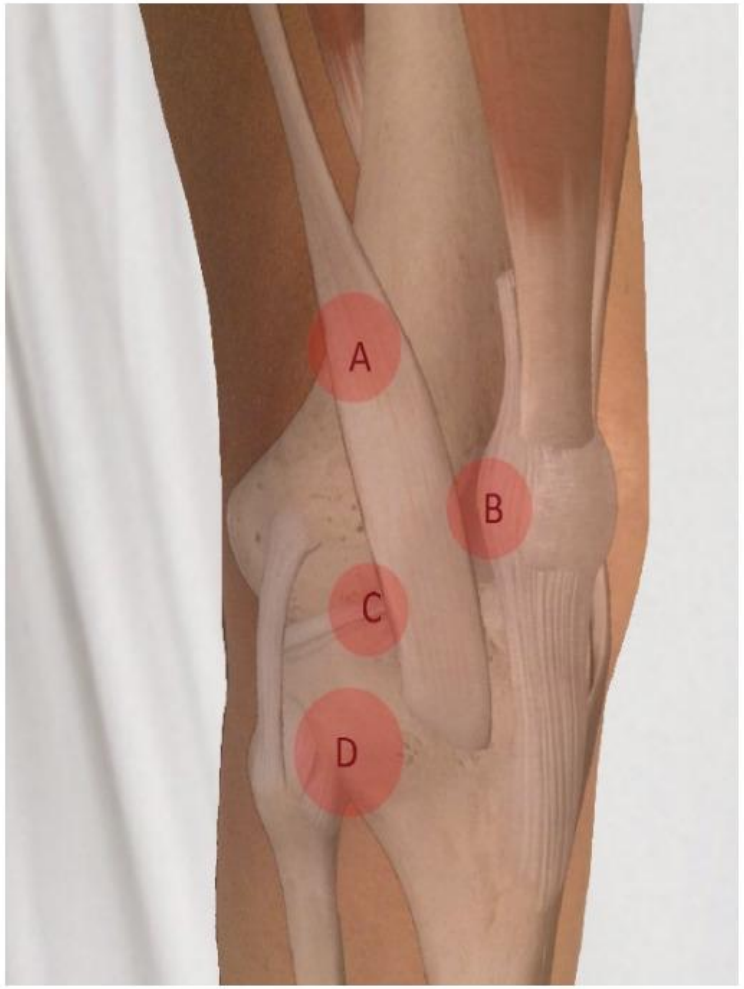


Figure 3 – Lateral Column



Questions

